

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP

#XUV 

ORGANIC CHEMISTRY:

SOME BASIC PRINCIPLES AND TECHNIQUES

LECTURE-01

CHEMISTRY

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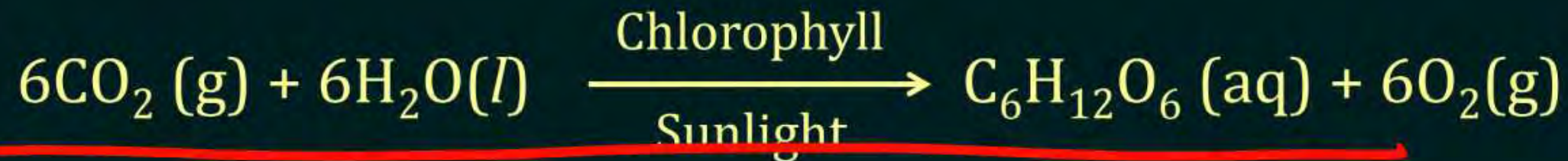
Topics To Covered

- 1 Vital Force Theory**
- 2 Tetravalency of Carbon**
- 3 Hybridisation**
- 4 Structural Representation of Organic Compounds**

Application of Redox Reactions

1. Photosynthesis.

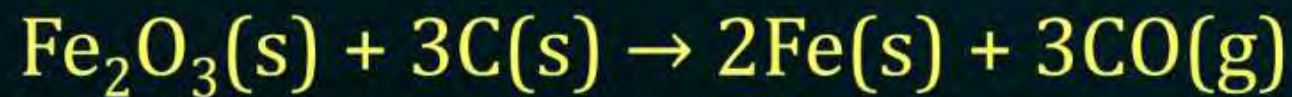
Photosynthesis is used to convert carbon dioxide and water by chlorophyll of green plants in the presence of sunlight to carbohydrates.



In this case, CO_2 is reduced to carbohydrates and water is oxidised to oxygen.

2. In metallurgy.

Many metal oxides are reduced to metals by using suitable reducing agents. For example, Fe_2O_3 is reduced to iron in a blast furnace using coke.



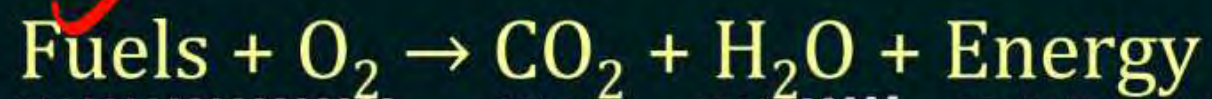
Application of Redox Reactions

3. Production of chemicals.

Many chemical compounds of our common need are commercially prepared by redox reactions. For example, sodium hydroxide, Cl_2 , F_2 , KOH , etc. are produced by electrolysis which is based on redox reactions.

4. Oxidation of fuels.

Oxidation of fuels is an important source of energy which satisfies our daily need of life. These are used for obtaining energy for domestic, transport or industrial purposes. They are also used for digestion of food in animals.



Application of Redox Reactions

5. Electrochemical cells or batteries.

The electrochemical cells or batteries are most important and are contributing significantly in different fields. Our vehicles cars, trucks, buses, etc. are moving on the storage cells based on redox reactions.

6. Quantitative relationships and stoichiometry.

Redox reactions also play significant role in quantitative analysis of different species by redox titrations. These titrations involve the reactions between oxidising and reducing agents and help to calculate the amount of unknown species in solutions.

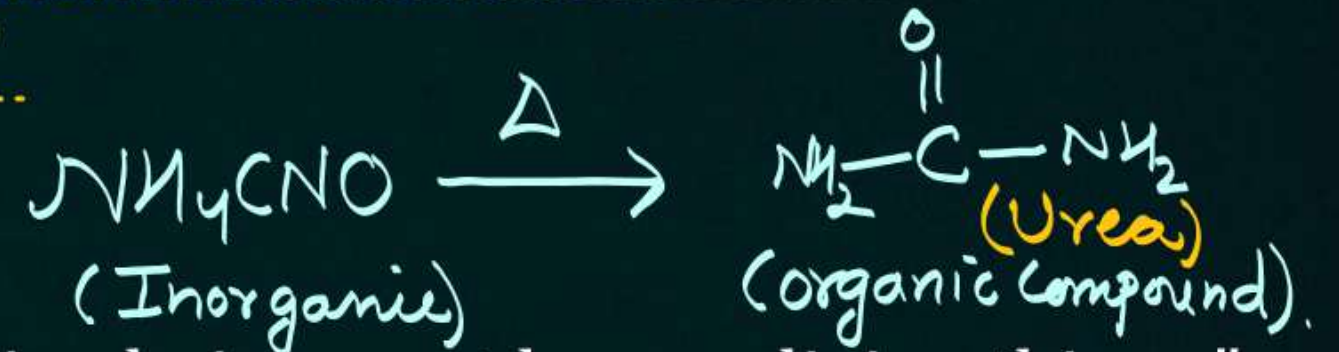
- 1) Photosynthesis
- 2) Metallurgy
- 3) Chemicals
- 4) Oxidation of fuels.
- 5) Electrochemical cell.
- 6) Quantitative analysis.

Lets start Organic Chemistry

ORGANIC CHEMISTRY:→

It is defined as branch of chemistry that deals with the study of hydrocarbons (compounds of carbon and hydrogen) and their derivatives.

- ## According to this theory:

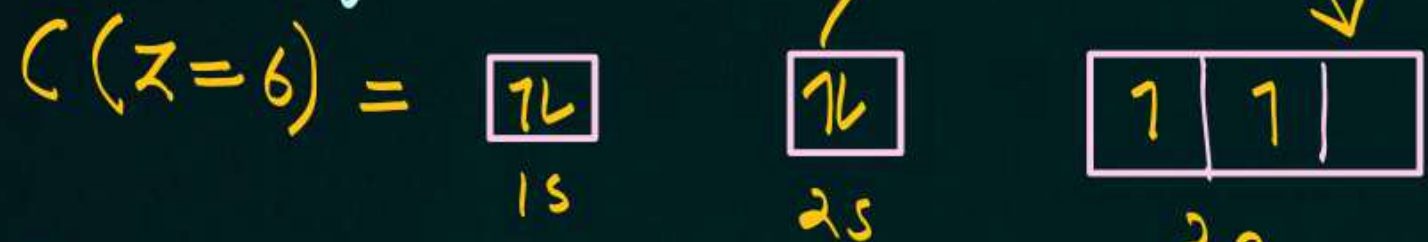


- "Organic compounds can be prepared only by living beings, not by non-living things."
- But later, this idea was **proved wrong** in 1828 when **Friedrich Wöhler** made **urea (an organic compound)** in the lab from **ammonium cyanate (an inorganic compound)** - without any 'vital force'.
- So, the **Vital Force Theory** was rejected, and it was proved that **organic compounds can be made in the laboratory too.**

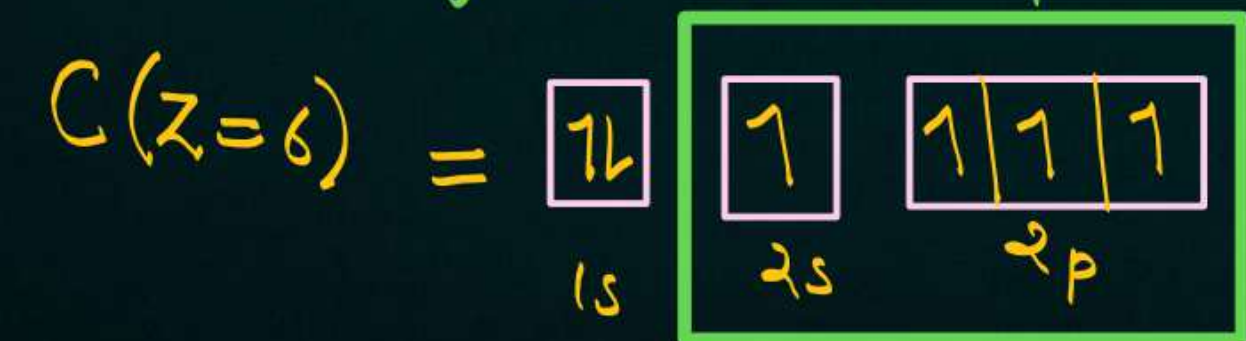
Tetravalency of Carbon



Carbon forms four bond i.e it is tetravalent.
We know that, ground state electronic configuration of carbon $\rightarrow$



In ground state it can form only two bond.
therefore during hybridisation promotion of e^- takes from $2s$ to $2p$.



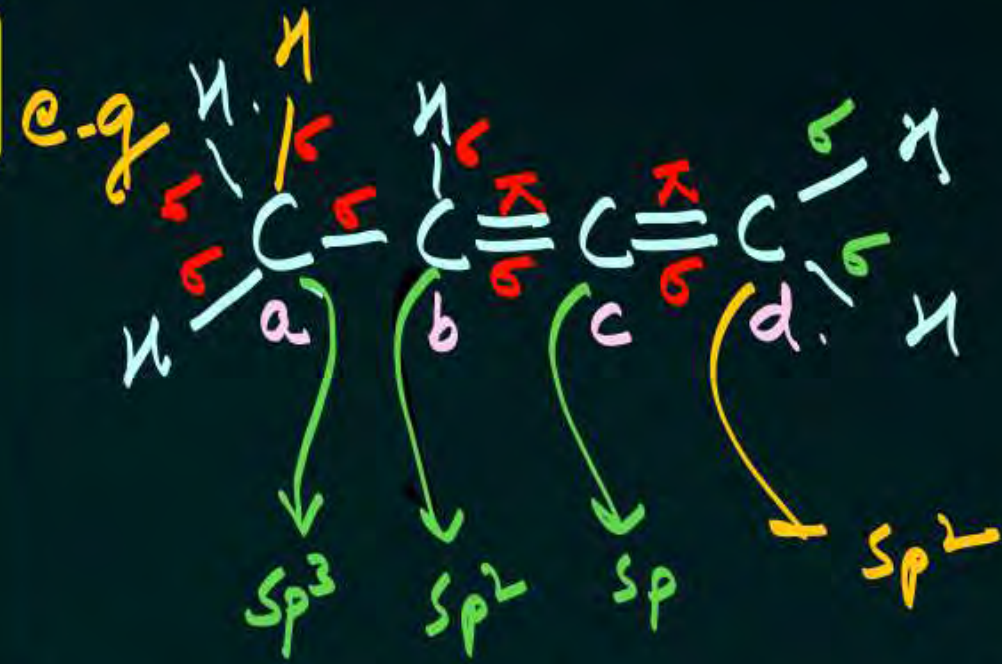
Therefore now 'C' has four unpaired e^- hence it can form four bonds.
& It can σ & π .

It can show sp , sp^2 & sp^3 hybridisation

Hybridisation (Determination)

CARBON.

No. of σ bonds	No. of π -bonds	Hybridisation
4	0	sp^3 (25% s + 75% p)
3	1	sp^2 (33.33% s + 66.66% p)
2	2	sp (50% s + 50% p)



%s d EN

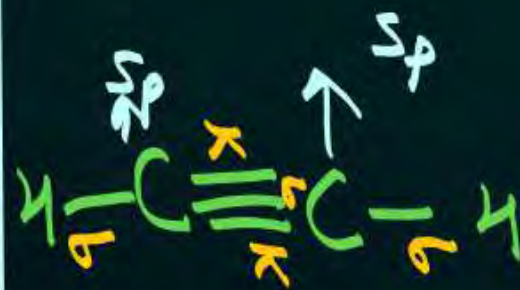
Percentage s character order.

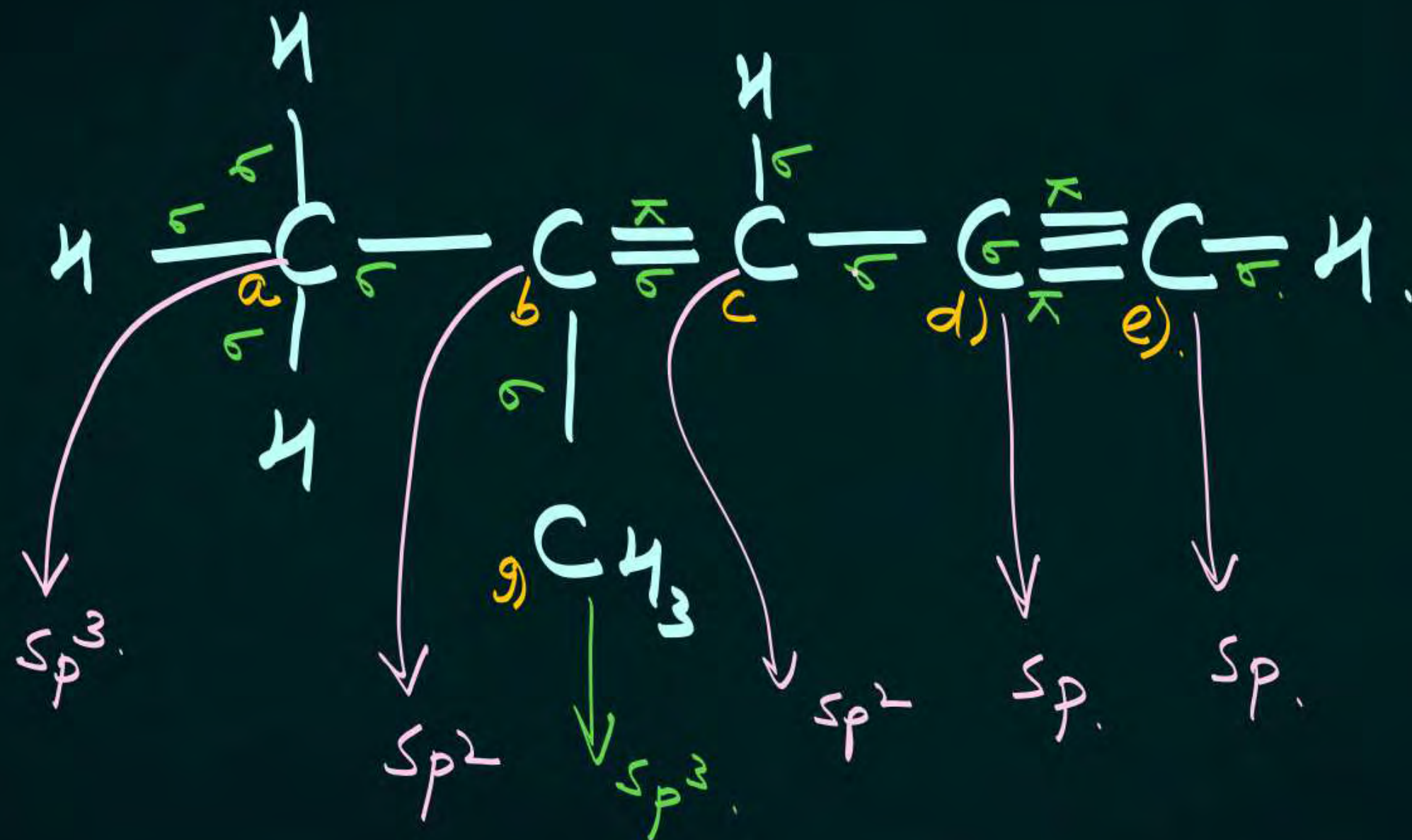
$$sp > sp^2 > sp^3$$

It has been found that more the percentage s character of carbon more is the tendency to attract shared pair of electron (electronegativity)

i.e.

$$C_{sp} > C_{sp^2} > C_{sp^3}$$





1. (S)

$Sp(S + P)$	50%
$Sp^2(S + P + P)$ 33.33 33.33 33.33	33.33% (S)
$Sp^3(S + P + P + P)$ 25 25 25 25	25%

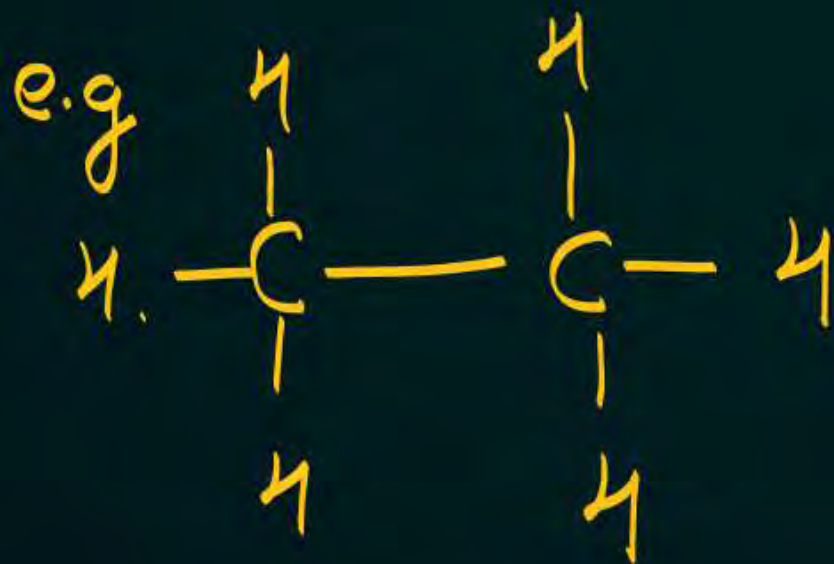
$$\frac{s}{s} \left(\frac{\frac{\pi}{s}}{\frac{s}{\pi}} \right) \left(\frac{s}{1} \right)$$

Structural Representation of Organic Compounds



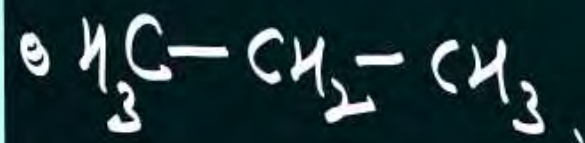
1) OPEN CHAIN STRUCTURE.

Each bond of a organic molecule is shown.



CONDENSED FORMULA.

- Bond between C & H is not shown & bond between C & C may or may not be shown.
- Branching is shown in brackets.



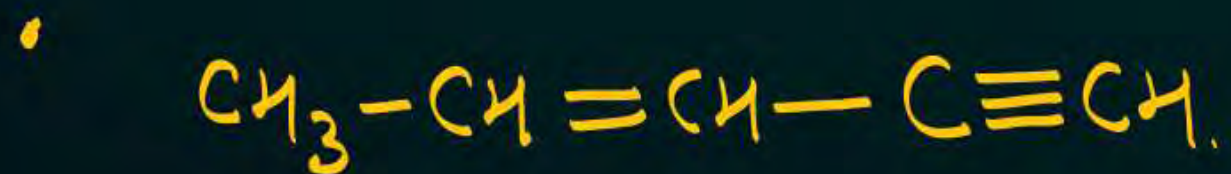
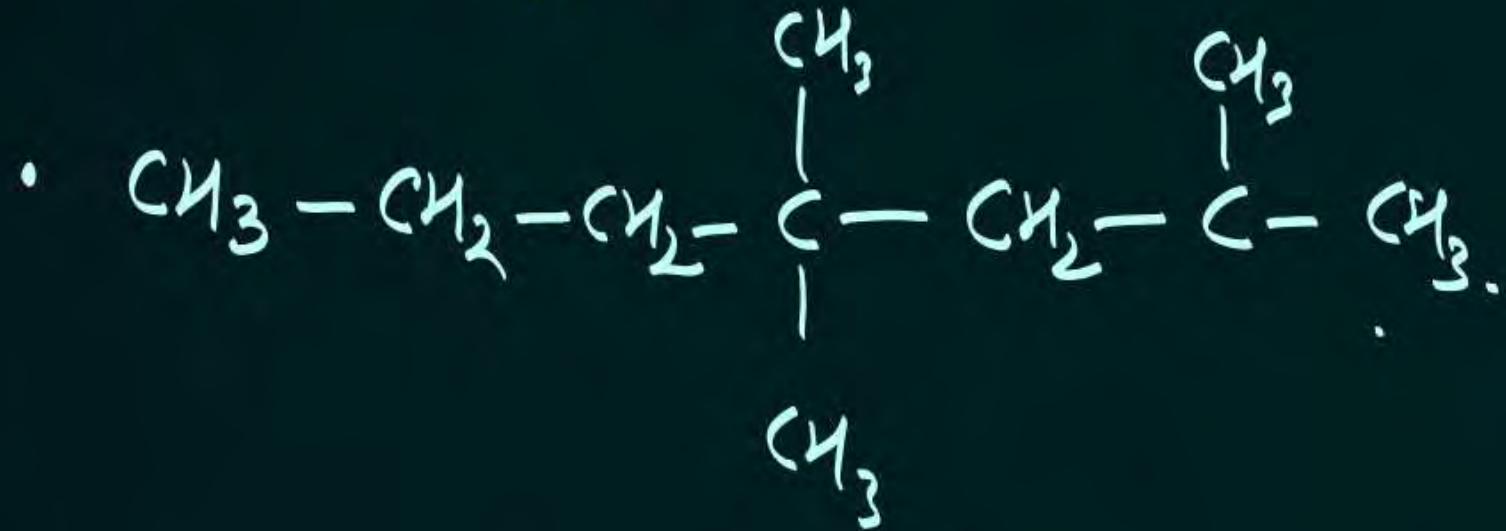
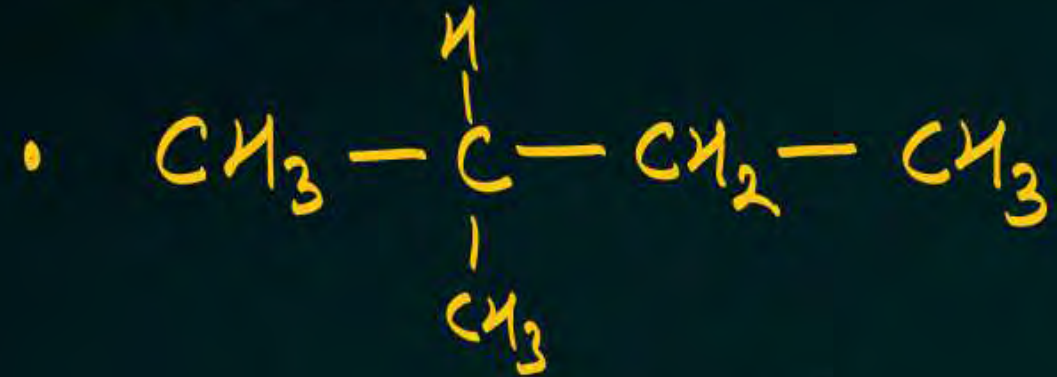
BOND line Notation.

- Each carbon attached with hydrogen is represented with the help of dot. and hydrogen attached with carbon is not shown. Other elements if present are shown as is.
- Carbon chain is shown in zig-zag format.



Structural Representation of Organic Compounds

Condensed.



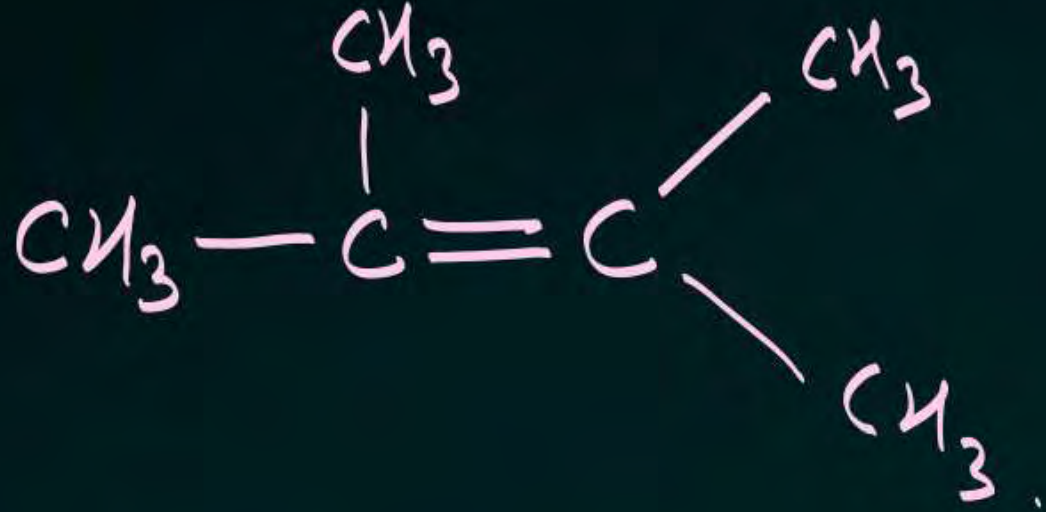
Bond line notation.



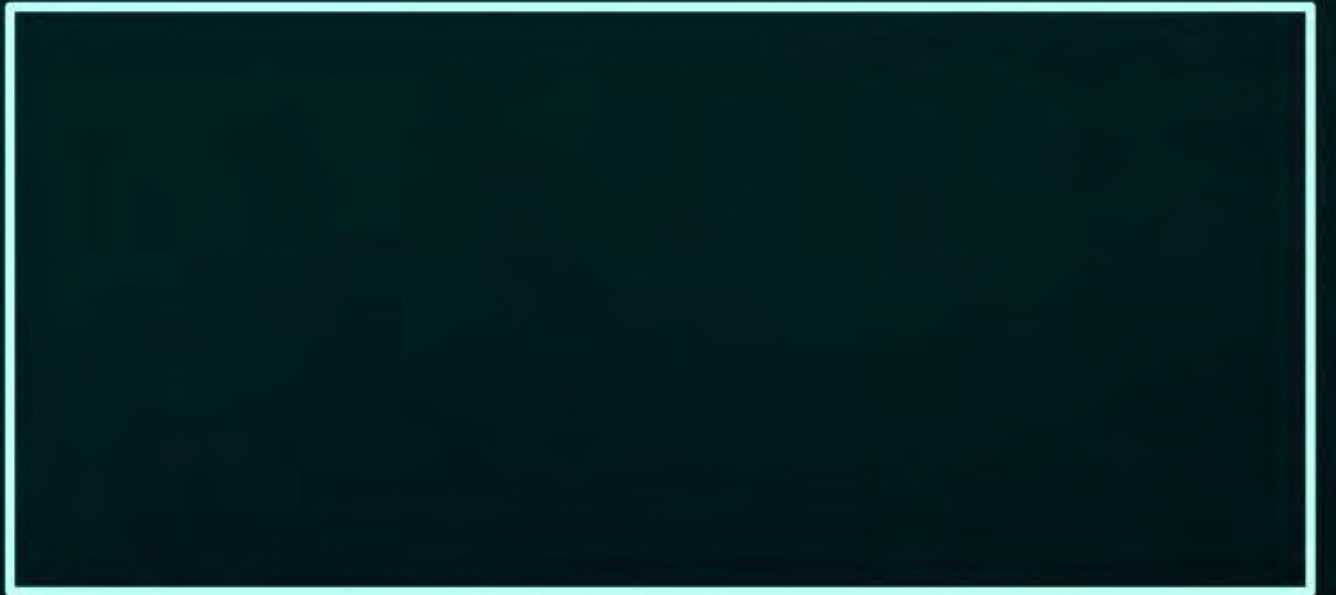
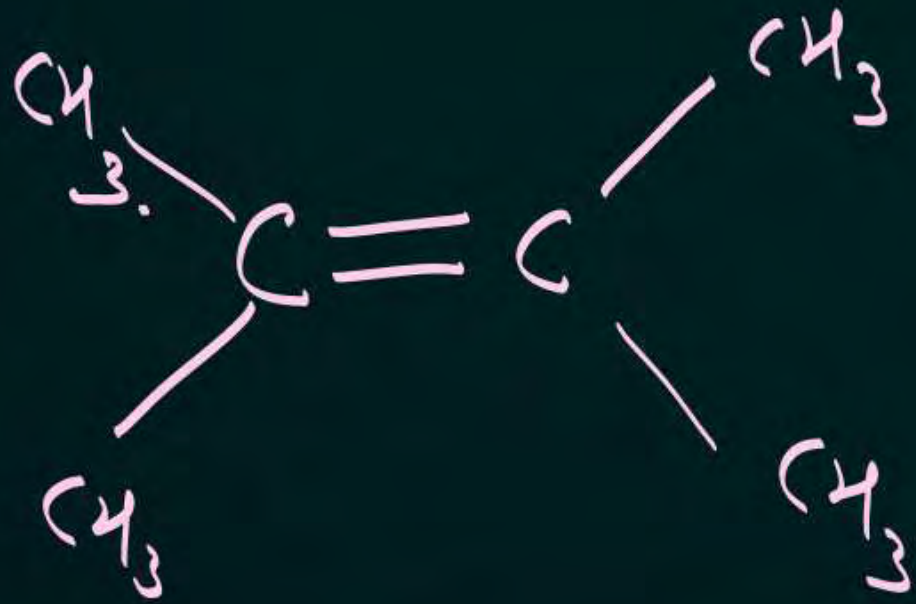
sp always linear

H.W

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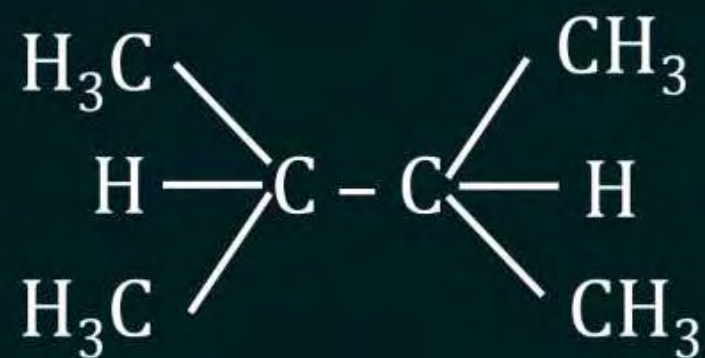


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QUESTION

Choose the correct bond-line formula for



QUESTION

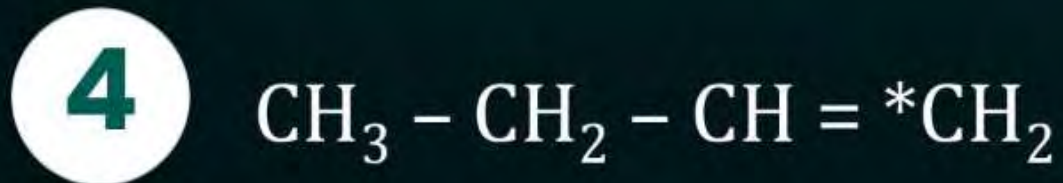


The correct order of electronegativity of carbon in ethane, ethene and ethyne is.

- 1 ethane < ethene < ethyne
- 2 ethyne < ethene < ethane
- 3 ethene < ethyne < ethane
- 4 ethene < ethane < ethyne

QUESTION

Electronegativity of carbon atoms depends upon their state of hybridisation. In which of the following compounds, the carbon marked with asterisk is most electronegative?



Thank
You